

Unit 3

Digital Markets

Framework Library · 20 reference sheets

3.1**Market Structures**

5 frameworks · pages 2 – 6

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3.4**Market Failures & Regulation**

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LEGEND

DEFINITION	Conceptual definitions and key terms (8 cards)
FRAMEWORK	Strategic frameworks and analytical tools (12 cards)
K	

Winner-Takes-All Markets

SLIBRARY 09 · SHEET
1 / 5

Why a single firm captures most of the value in many digital markets.

PROFESSIONAL DEFINITION

A market structure in which network effects, increasing returns to scale, and switching costs converge so that one or two firms acquire dominant share, while smaller competitors are squeezed out. Common in digital markets where marginal cost is near zero and connection density compounds.

WHO DEVELOPED IT

Frank & Cook (1995) 'The Winner-Take-All Society' (free press); refined for digital markets by Eisenmann, Parker & Van Alstyne (HBR, 2006).

WHY IT WAS DEVELOPED

Brian Arthur's earlier work on increasing returns in technology (Arthur, 1989) provides the theoretical backbone.

FIGURE 1 · WINNER-TAKES-ALL MARKETS



Figure 1. Winner-Takes-All Markets: Why a single firm captures most of the value in many digital markets.

HOW IT IS USED

when forecasting whether a fragmented digital market will consolidate.

WHEN IT IS USED

in regulatory analysis — diagnoses which markets need ex-ante intervention (DMA logic).

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Structures.

REAL-WORLD EXAMPLES

● Search: Google ≈ 90% global share for two decades.

● Desktop OS: Microsoft Windows ≈ 73% global share.

● Social: Meta family of apps reaches ≈ 4 bn users monthly (2024).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Winner-Takes-All Markets contributes a clearer, more transferable framing of the dynamic it names — Why a single firm captures most of the value in many digital markets. Its analytical contribution is the explicit recognition that three conditions converge: (i) strong network effects, (ii) high multi-homing costs, (iii) limited niche differentiation.

RELATED FRAMEWORKS IN THIS COURSE

Two-Sided Markets — Market Structure View · 3.1.2

GAFAM Concentration · 3.1.5

DMA — Digital Markets Act · 3.4.1

SOURCES (HARVARD REFERENCING STYLE)

Arthur, W.B. (1989) 'Competing technologies, increasing returns, and lock-in by historical events', *The Economic Journal*, 99(394), pp. 116–131.

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', *Harvard Business Review*, 84(10), pp. 92–101.

Frank, R.H. and Cook, P.J. (1995) *The Winner-Take-All Society*. New York: Free Press.

Two-Sided Markets — Market Structure View

SLIBRARY 09 · SHEET
2 / 5

Rochet & Tirole — markets where the platform itself is the unit of competition.

PROFESSIONAL DEFINITION

A market structure in which a platform mediates transactions between two distinct user groups whose value to each other depends on each side's participation. Unlike conventional markets, prices on each side are set jointly to optimise platform-level performance, not side-level marginal cost.

WHO DEVELOPED IT

Rochet, J.-C. and Tirole, J. (2003) 'Platform Competition in Two-Sided Markets', Journal of the European Economic Association.

WHY IT WAS DEVELOPED

Tirole's 2014 Nobel Prize cited this work. Earlier intuition in credit-card economics (Baxter, 1983).

FIGURE 1 · TWO-SIDED MARKETS — MARKET STRUCTURE VIEW



Figure 1. Two-Sided Markets — Market Structure View: Rochet & Tirole — markets where the platform itself is the unit of competition.

HOW IT IS USED

when analysing competition in any platform market — the unit of analysis is the platform, not the firm.

WHEN IT IS USED

in competition policy — explains why traditional antitrust tools (price-cost margins) often misdiagnose digital markets.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Structures.

REAL-WORLD EXAMPLES

● Credit cards: merchants vs. consumers — Visa/Mastercard set differential pricing.

● Operating systems: developers vs. users — Microsoft, Apple, Google compete on both fronts.

● Job platforms: recruiters vs. candidates — LinkedIn vs. Indeed compete on combined network density.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Two-Sided Markets — Market Structure View contributes a clearer, more transferable framing of the dynamic it names — Rochet & Tirole — markets where the platform itself is the unit of competition. Its analytical contribution is the explicit recognition that two user groups (A and B); each side's utility increases when the other side is larger (cross-side network effects).

RELATED FRAMEWORKS IN THIS COURSE

[Multi-Sided Platforms · 3.1.3](#)
[Subsidised Side Pricing · 3.3.2](#)
[Direct vs Indirect Network Effects \(Unit 2.2.2\)](#)

SOURCES (HARVARD REFERENCING STYLE)

Rochet, J.-C. and Tirole, J. (2003) 'Platform Competition in Two-Sided Markets', Journal of the European Economic Association, 1(4), pp. 990–1029.

Rochet, J.-C. and Tirole, J. (2006) 'Two-sided markets: a progress report', RAND Journal of Economics, 37(3), pp. 645–667.

Evans, D.S. and Schmalensee, R. (2016) Matchmakers: The New Economics of Multisided Platforms. Boston: Harvard Business Review Press.

Multi-Sided Platforms

SLIBRARY 09 · SHEET
3 / 5

Hagiu & Wright — three or more interdependent user groups.

PROFESSIONAL DEFINITION

A multi-sided platform (MSP) enables direct interactions between two or more distinct types of customers, each affiliated with the platform. The defining feature is direct interaction (transaction or content exchange) plus affiliation by multiple customer groups.

WHO DEVELOPED IT

Conceptual foundations in Rochet & Tirole (2003); refined by Hagiu, A. and Wright, J. (2015) 'Multi-Sided Platforms', International Journal of Industrial Organization, 43, pp. 162–174.

WHY IT WAS DEVELOPED

Parker, Van Alstyne & Choudary (2016) 'Platform Revolution' provides the strategic playbook.

FIGURE 1 · MULTI-SIDED PLATFORMS

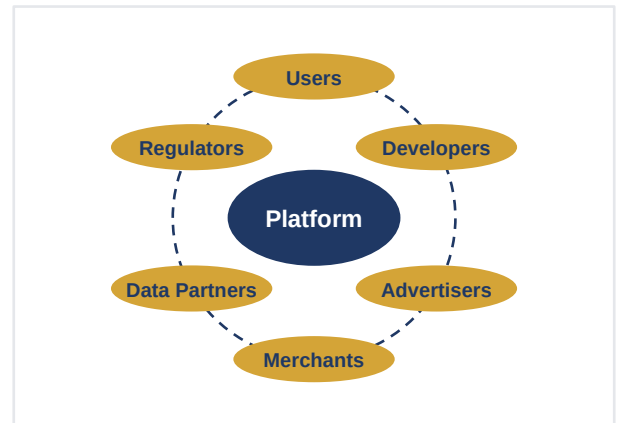


Figure 1. Multi-Sided Platforms: Hagiu & Wright — three or more interdependent user groups.

HOW IT IS USED

when designing a platform business model — forces clarity on which sides the platform serves.

WHEN IT IS USED

in M&A and competition diligence — multi-sidedness is the source of platform valuation premium.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Structures.

REAL-WORLD EXAMPLES

● Apple App Store: users + developers + Apple (curator) + advertisers.

● Google Search: users + advertisers + content publishers + AI training data partners.

● Amazon: shoppers + sellers + Amazon retail + advertisers + AWS clients + logistics partners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Multi-Sided Platforms contributes a clearer, more transferable framing of the dynamic it names — Hagiu & Wright — three or more interdependent user groups. Its analytical contribution is the explicit recognition that three or more sides — users, complementors, developers, advertisers, data partners — each with distinct value propositions.

RELATED FRAMEWORKS IN THIS COURSE

Two-Sided Markets — Market Structure View · 3.1.2

Platform Governance · 3.3.3

API Economy (Unit 2.1.4)

SOURCES (HARVARD REFERENCING STYLE)

Hagiu, A. and Wright, J. (2015) 'Multi-sided platforms', International Journal of Industrial Organization, 43, pp. 162–174.

Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York: W.W. Norton.

Gawer, A. (2014) 'Bridging different perspectives on technological platforms: toward an integrative framework', Research Policy, 43(7), pp. 1239–1249.

Marketplace vs. Platform

SLIBRARY 09 · SHEET
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All marketplaces are platforms; not all platforms are marketplaces.

PROFESSIONAL DEFINITION

A marketplace is a digital venue where transactions are completed between independent buyers and sellers. A platform is the broader category — any digital architecture mediating interactions between distinct user groups, with or without direct transactions. The distinction matters strategically and legally.

WHO DEVELOPED IT

Marketplace formalised by Bakos, Y. (1998) 'The emerging role of electronic marketplaces on the Internet', Communications of the ACM.

WHY IT WAS DEVELOPED

Platform definition refined by Gawer (2014) and Parker et al. (2016). EU regulation (P2B Regulation 2019, DSA 2022) operationalises the distinction.

FIGURE 1 · MARKETPLACE VS. PLATFORM



Figure 1. Marketplace vs. Platform: All marketplaces are platforms; not all platforms are marketplaces.

HOW IT IS USED

when scoping a new digital business — clarifies whether the offering is transactional or interactional.

WHEN IT IS USED

in regulatory mapping — different rule-sets apply to each.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Market Structures.

REAL-WORLD EXAMPLES

● Marketplaces: Amazon, eBay, Vinted, Booking.com — transactions are central.

● Platforms (non-marketplace): Facebook, X/Twitter, TikTok — interaction is central, transactions optional.

● Hybrids: Instagram (platform with shopping marketplace), YouTube (platform with creator monetisation).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Marketplace vs. Platform contributes a clearer, more transferable framing of the dynamic it names — All marketplaces are platforms; not all platforms are marketplaces. Its analytical contribution is the explicit recognition that marketplace: intermediates a discrete transaction (good, service, content) — Amazon Marketplace, eBay, Etsy.

RELATED FRAMEWORKS IN THIS COURSE

[Multi-Sided Platforms · 3.1.3](#)
[DSA · 3.4.2](#)
[Pipeline vs Platform](#)

SOURCES (HARVARD REFERENCING STYLE)

Bakos, Y. (1998) 'The emerging role of electronic marketplaces on the Internet', Communications of the ACM, 41(8), pp. 35–42.

European Union (2019) Regulation (EU) 2019/1150 on Promoting Fairness and Transparency for Business Users of Online Intermediation Services. Brussels: Official Journal of the EU.

Gawer, A. (2014) 'Bridging different perspectives on technological platforms: toward an integrative framework', Research Policy, 43(7), pp. 1239–1249.

GAFAM Concentration

Five firms · disproportionate share of digital market value.

PROFESSIONAL DEFINITION

The empirical pattern of market concentration among Google (Alphabet), Apple, Facebook (Meta), Amazon, and Microsoft — five US-headquartered firms that together account for a dominant share of global digital advertising, app distribution, cloud, and consumer software markets. UNCTAD (WIR 2025) reports the top 5 digital MNEs hold ~48% of revenue across the top 100 digital firms.

WHO DEVELOPED IT

Term popularised in European policy discourse around 2017–2018.

WHY IT WAS DEVELOPED

UNCTAD World Investment Report tracks digital MNE concentration annually. Lina Khan's 'Amazon's Antitrust Paradox' (2017, Yale Law Journal) reframed the academic debate.

FIGURE 1 · GAFAM CONCENTRATION

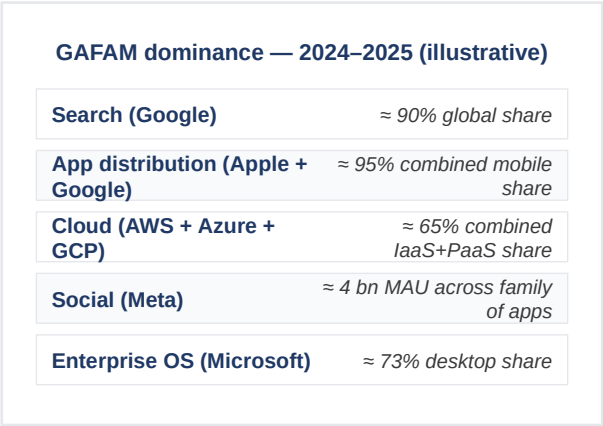


Figure 1. GAFAM Concentration: Five firms · disproportionate share of digital market value.

HOW IT IS USED

when assessing competitive dynamics in any digital market — GAFAM is often a present or potential entrant.

WHEN IT IS USED

in regulatory and policy analysis — frames the rationale for the EU DMA.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Structures.

REAL-WORLD EXAMPLES

- Google: 90% search · 70% mobile OS (Android) · 30% cloud · 10% smart speakers.
- Apple: 60% US smartphone share · 50% global app store revenue · \$400bn+ services revenue.
- Microsoft: 73% desktop OS · 25% cloud · 80% productivity (Office) · OpenAI partnership.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, GAFAM Concentration contributes a clearer, more transferable framing of the dynamic it names — Five firms · disproportionate share of digital market value. Its analytical contribution is the explicit recognition that each firm controls one or more 'gatekeeper' positions: search (Google), app distribution (Apple, Google), cloud (Amazon, Microsoft, Google), social (Meta), enterprise software (Microsoft).

RELATED FRAMEWORKS IN THIS COURSE

- Winner-Takes-All Markets · 3.1.1
- DMA · 3.4.1
- Antitrust in Digital Markets · 3.4.3

SOURCES (HARVARD REFERENCING STYLE)

Khan, L. (2017) 'Amazon's antitrust paradox', Yale Law Journal, 126(3), pp. 710–805.

UNCTAD (2025) World Investment Report 2025: International Investment in the Digital Economy. Geneva: United Nations.

European Commission (2024) Designating Gatekeepers under the Digital Markets Act. Brussels: European Commission.

Fintech

Technology-enabled innovation in financial services — payments, lending, wealth, insurance.

PROFESSIONAL DEFINITION

Fintech denotes the application of digital technologies — APIs, mobile, cloud, AI, distributed ledgers — to deliver financial services more efficiently, accessibly, or in entirely new forms. Encompasses payments, lending, wealth management, insurance technology, capital markets, and regulatory technology.

WHO DEVELOPED IT

Term in use since the early 2010s; conceptual roots in Arner, Barberis and Buckley (2016) 'The Evolution of Fintech: A New Post-Crisis Paradigm' — University of Hong Kong.

WHY IT WAS DEVELOPED

Industry growth catalysed post-2008 financial crisis as incumbents retrenched and new entrants exploited digital channels.

FIGURE 1 · FINTECH

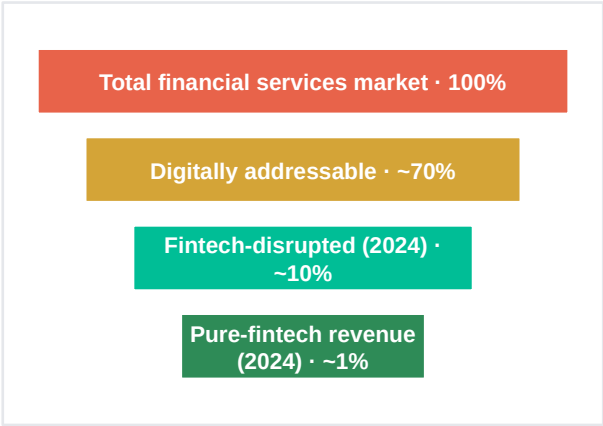


Figure 1. Fintech: Technology-enabled innovation in financial services — payments, lending, wealth, insurance.

HOW IT IS USED

when scoping the addressable market for any new digital financial product.

WHEN IT IS USED

as a backdrop to Unit 3 case studies on platform competition in finance.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Verticals.

REAL-WORLD EXAMPLES

- Stripe: API-first payments — ~\$1.4T processed in 2024.
- Revolut: neobank — 50m+ users globally (2025).
- Klarna: BNPL — 150m+ active users; IPO 2025.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Fintech contributes a clearer, more transferable framing of the dynamic it names — Technology-enabled innovation in financial services — payments, lending, wealth, insurance. Its analytical contribution is the explicit recognition that five core verticals: Payments, Lending, Wealth Management, InsurTech, RegTech.

RELATED FRAMEWORKS IN THIS COURSE

- Insurtech · 3.2.3
- API Economy (Unit 2.1.4)
- Platform Governance · 3.3.3

SOURCES (HARVARD REFERENCING STYLE)

Arner, D., Barberis, J. and Buckley, R. (2016) 'The evolution of FinTech: a new post-crisis paradigm?', Georgetown Journal of International Law, 47, pp. 1271–1319.

BCG (2024) Global Fintech Report 2024. Boston: Boston Consulting Group.

McKinsey & Company (2024) Global Banking Annual Review 2024. New York: McKinsey.

Legaltech

SLIBRARY 10 · SHEET
2 / 5

Digital tools transforming legal practice and access to justice.

PROFESSIONAL DEFINITION

Legaltech refers to the application of technology — document automation, e-discovery, contract analytics, AI legal research, online dispute resolution — to legal practice and the delivery of legal services. Encompasses tools used by law firms, in-house counsel, courts, and direct-to-consumer legal platforms.

WHO DEVELOPED IT

Susskind, R. (2010) 'The End of Lawyers?' and 'Tomorrow's Lawyers' (2013, 2017) provided the conceptual frame.

WHY IT WAS DEVELOPED

Earlier roots in case-management software (1990s); modern AI wave from ~2017 with neural-network legal research.

FIGURE 1 · LEGALTECH



Figure 1. Legaltech: Digital tools transforming legal practice and access to justice.

HOW IT IS USED

when assessing how any professional-services industry digitises — legal is a useful template.

WHEN IT IS USED

in case studies of platform disruption in regulated industries.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Verticals.

REAL-WORLD EXAMPLES

● Harvey AI (2023+): GPT-based legal-research assistant; backed by OpenAI Startup Fund.

● DoNotPay: consumer legal automation — speeding-ticket disputes, contract cancellation.

● Relativity: e-discovery software; used in major litigation worldwide.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Legaltech contributes a clearer, more transferable framing of the dynamic it names — Digital tools transforming legal practice and access to justice. Its analytical contribution is the explicit recognition that four primary application areas: (i) document automation and contract drafting; (ii) e-discovery and litigation analytics; (iii) AI-powered legal research; (iv) online dispute resolution and self-service legal forms.

RELATED FRAMEWORKS IN THIS COURSE

Insurtech · 3.2.3

AI & Copyright (Unit 5)

DSA — Digital Services Act · 3.4.2

SOURCES (HARVARD REFERENCING STYLE)

Susskind, R. (2017) *Tomorrow's Lawyers: An Introduction to Your Future*. 2nd edn. Oxford: Oxford University Press.

Thomson Reuters (2024) *Future of Professionals Report*. Eagan, MN: Thomson Reuters.

Sourdin, T. (2018) 'Judge v. robot: artificial intelligence and judicial decision-making', *UNSW Law Journal*, 41(4), pp. 1114–1133.

Insurtech

SLIBRARY 10 · SHEET
3 / 5

Digital transformation of underwriting, claims, distribution, and risk pricing.

PROFESSIONAL DEFINITION

Insurtech is the application of digital technologies — IoT sensors, AI risk-pricing, mobile distribution, parametric contracts, and embedded insurance — to redesign the insurance value chain. Spans auto, home, life, health, commercial, and parametric (climate, cyber) coverage.

WHO DEVELOPED IT

Term emerged 2015–2016 alongside the broader fintech wave.

WHY IT WAS DEVELOPED

Conceptual roots in IoT and big-data underwriting (Eling and Lehmann, 2018, Geneva Papers). Key early ventures: Lemonade (2015, US), ZhongAn (2013, China), Trōv (2012).

FIGURE 1 · INSURTECH

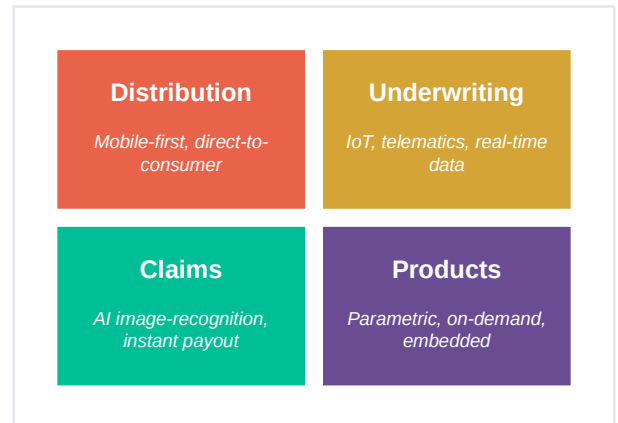


Figure 1. Insurtech: Digital transformation of underwriting, claims, distribution, and risk pricing.

HOW IT IS USED

when assessing platform disruption in any sector with regulated risk-pooling.

WHEN IT IS USED

as a case study of AI in customer-facing financial services.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Verticals.

REAL-WORLD EXAMPLES

- Lemonade: AI-first home & renters insurance — public 2020.
- ZhongAn: Tencent/Alibaba/Ping An joint venture — 500m+ policies.
- Root Insurance: telematics-priced auto cover (US).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Insurtech contributes a clearer, more transferable framing of the dynamic it names — Digital transformation of underwriting, claims, distribution, and risk pricing. Its analytical contribution is the explicit recognition that four application zones: (i) distribution — direct-to-consumer mobile; (ii) underwriting — IoT-driven risk pricing; (iii) claims — AI image recognition for damage assessment; (iv) products — parametric and on-demand cover.

RELATED FRAMEWORKS IN THIS COURSE

Fintech · 3.2.1

Data as Strategic Asset (Unit 2.3.4)

GDPR Principles (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Eling, M. and Lehmann, M. (2018) 'The impact of digitalization on the insurance value chain and the insurability of risks', Geneva Papers on Risk and Insurance, 43(3), pp. 359–396.

OECD (2020) The Impact of Big Data and Artificial Intelligence (AI) in the Insurance Sector. Paris: OECD Publishing.

Cappiello, A. (2020) The Technological Disruption of Insurance Industry: A Methodological Investigation. Cham: Springer.

Regtech

SLIBRARY 10 · SHEET
4 / 5

Technology for regulatory compliance — automating the cost of being a regulated firm.

PROFESSIONAL DEFINITION

Regulatory technology — automation of compliance, monitoring, reporting, and risk management for firms operating in regulated environments. Spans KYC/AML in finance, GDPR/data-protection compliance, ESG reporting, AI-Act compliance, and emerging cybersecurity-mandate (NIS2) tooling.

WHO DEVELOPED IT

Arner, D., Barberis, J. and Buckley, R. (2017) 'FinTech, RegTech, and the Reconceptualization of Financial Regulation', Northwestern Journal of International Law and Business.

WHY IT WAS DEVELOPED

Industry term predates academic naming — used in UK Financial Conduct Authority sandbox (2015).

FIGURE 1 · REGTECH

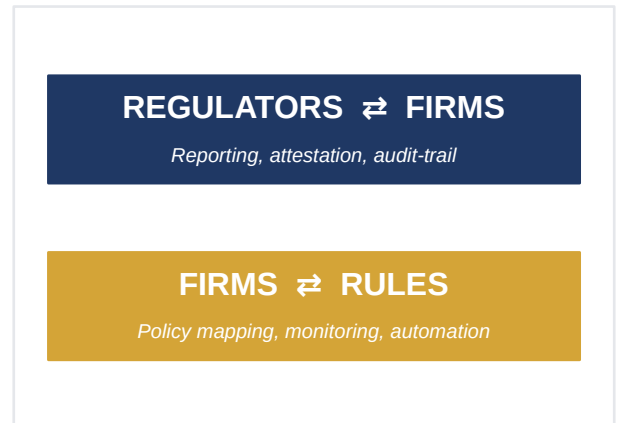


Figure 1. Regtech: Technology for regulatory compliance — automating the cost of being a regulated firm.

HOW IT IS USED

when scoping the compliance overhead of any regulated digital business.

WHEN IT IS USED

as a case study of how regulation creates new platform-business opportunities.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Verticals.

REAL-WORLD EXAMPLES

- ComplyAdvantage: AI-driven AML / sanctions screening.
- Onfido: identity-verification (KYC) APIs.
- Workiva: SEC and ESG reporting platform.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Regtech contributes a clearer, more transferable framing of the dynamic it names — Technology for regulatory compliance — automating the cost of being a regulated firm. Its analytical contribution is the explicit recognition that four application areas: (i) regulatory reporting; (ii) KYC / AML / sanctions screening; (iii) transaction monitoring and fraud detection; (iv) policy management and audit trail.

RELATED FRAMEWORKS IN THIS COURSE

Fintech · 3.2.1

DMA / DSA (3.4.1, 3.4.2)

EU AI Act (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Arner, D., Barberis, J. and Buckley, R. (2017) 'FinTech, RegTech, and the reconceptualization of financial regulation', Northwestern Journal of International Law and Business, 37(3), pp. 371–413.

Financial Stability Board (2020) The Use of Supervisory and Regulatory Technology by Authorities and Regulated Institutions. Basel: FSB.

Deloitte (2024) RegTech Universe 2024. London: Deloitte.

Proptech

SLIBRARY 10 · SHEET
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Property technology — digitising real-estate transactions, operations, and tenant experience.

PROFESSIONAL DEFINITION

Proptech (property technology) covers digital innovations across the real-estate lifecycle: search and discovery, transaction (buy / rent / sell), financing, construction-tech, building operations and IoT, and tenant-experience platforms. Spans residential and commercial, owner-occupied and investor markets.

WHO DEVELOPED IT

Term in use since ~2014 in UK and US venture press; conceptual lineage in Saull and Baum (2017) 'PropTech 3.0: The future of real estate' — University of Oxford / Saïd Business School.

WHY IT WAS DEVELOPED

The framework was developed to give practitioners a precise, transferable vocabulary for the dynamic captured in its tagline — replacing the looser pre-existing terms used in earlier industry and academic discourse.

FIGURE 1 · PROPTech

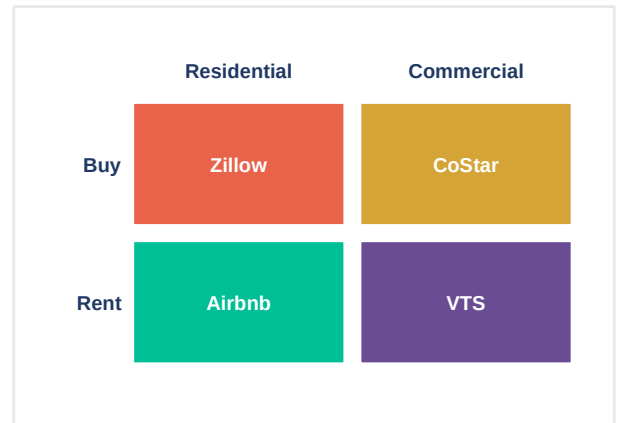


Figure 1. Proptech: Property technology — digitising real-estate transactions, operations, and tenant experience.

HOW IT IS USED

when assessing platform disruption in capital-intensive, regulated, high-trust industries.

WHEN IT IS USED

as a counterpoint to fintech — proptech demonstrates how digitisation slows in offline-anchored sectors.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Verticals.

REAL-WORLD EXAMPLES

● Zillow / Rightmove / ImmoScout24: residential listings and search.

● Airbnb: short-term rental marketplace — disrupted hospitality and zoning.

● Matterport: 3D building capture and digital twins.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Proptech contributes a clearer, more transferable framing of the dynamic it names — Property technology — digitising real-estate transactions, operations, and tenant experience. Its analytical contribution is the explicit recognition that three waves: 1.0 — listings and search portals (Zillow, Rightmove, ImmoScout); 2.0 — transaction platforms (Compass, OpenDoor); 3.0 — IoT, smart-building and AI-driven analytics (Matterport, VTS, CoStar).

RELATED FRAMEWORKS IN THIS COURSE

Fintech · 3.2.1

Data as Strategic Asset (Unit 2.3.4)

Marketplace vs Platform · 3.1.4

SOURCES (HARVARD REFERENCING STYLE)

Saull, A. and Baum, A. (2017) PropTech 3.0: The Future of Real Estate. Oxford: Saïd Business School, University of Oxford.

Baum, A. (2020) PropTech 2020: The Future of Real Estate. Oxford: Saïd Business School, University of Oxford.

Mordor Intelligence (2024) PropTech Market Size & Share Analysis 2024–2029. Hyderabad: Mordor Intelligence.

Chicken-and-Egg Problem

SLIBRARY 11 · SHEET
1 / 5

How does a two-sided platform launch when neither side will join without the other?

PROFESSIONAL DEFINITION

The platform-launch problem: with cross-side network effects, neither side will join in scale until the other side is present. The platform must therefore find a launch sequence that breaks this circularity.

WHO DEVELOPED IT

Caillaud, B. and Jullien, B. (2003) 'Chicken & Egg: Competition Among Intermediation Service Providers', RAND Journal of Economics.

WHY IT WAS DEVELOPED

Strategic playbook formalised by Eisenmann, Parker, Van Alstyne (2006) and Hagiu (2014).

FIGURE 1 · CHICKEN-AND-EGG PROBLEM



Figure 1. Chicken-and-Egg Problem: How does a two-sided platform launch when neither side will join without the other?.

HOW IT IS USED

at the inception of any new platform business — must be solved before product launch.

WHEN IT IS USED

in venture diligence — the launch playbook is a major valuation determinant.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Platform Economics.

REAL-WORLD EXAMPLES

● PayPal: bootstrapped on eBay — sellers wanted faster payment.

● Uber: aggressive driver-side subsidies before passenger scaling.

● OpenAI / ChatGPT: bypassed B2B by going direct to consumers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Chicken-and-Egg Problem contributes a clearer, more transferable framing of the dynamic it names — How does a two-sided platform launch when neither side will join without the other?. Its analytical contribution is the explicit recognition that four common solutions: (i) Subsidise one side — typically the side with stronger cross-side externalities; (ii) Single-side launch — bootstrap value within one side first (e.g. dating, then add events); (iii) 'Marquee' anchor — secure a high-profile partner whose presence draws the other side; (iv) Existing-network piggyback — leverage another network to bootstrap (e.g. PayPal on eBay).

RELATED FRAMEWORKS IN THIS COURSE

Two-Sided Markets — Market Structure View · 3.1.2

Subsidised Side Pricing · 3.3.2

Connection Density (Unit 2.2.4)

SOURCES (HARVARD REFERENCING STYLE)

Caillaud, B. and Jullien, B. (2003) 'Chicken & egg: competition among intermediation service providers', RAND Journal of Economics, 34(2), pp. 309–328.

Eisenmann, T., Parker, G. and Van Alstyne, M. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92–101.

Hagiu, A. (2014) 'Strategic decisions for multisided platforms', MIT Sloan Management Review, 55(2), pp. 71–80.

Subsidised Side Pricing

SLIBRARY 11 · SHEET
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Charge one side · subsidise the other · maximise platform-level surplus.

PROFESSIONAL DEFINITION

On a two-sided platform, profit-maximising prices are set jointly across sides. The 'subsidy side' is typically priced below marginal cost (often free), while the 'money side' bears the platform's profit. The choice is driven by which side has stronger cross-side externalities, lower price-elasticity, or higher multi-homing costs.

WHO DEVELOPED IT

Rochet and Tirole (2003); refinement in Armstrong, M. (2006) 'Competition in Two-Sided Markets', RAND Journal of Economics.

WHY IT WAS DEVELOPED

Industry codification in Evans and Schmalensee (2016) 'Matchmakers'.

FIGURE 1 · SUBSIDISED SIDE PRICING

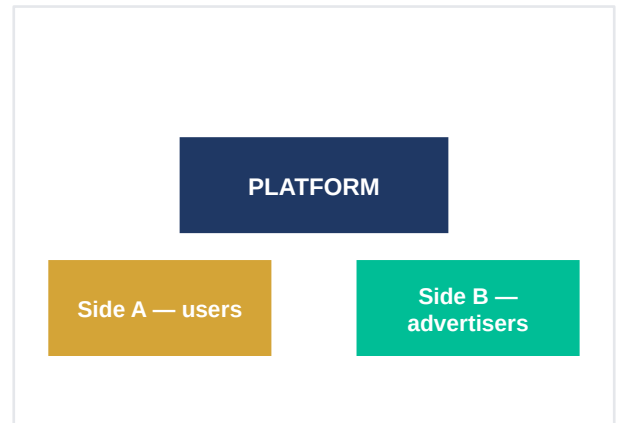


Figure 1. Subsidised Side Pricing: Charge one side · subsidise the other · maximise platform-level surplus.

HOW IT IS USED

when designing or auditing platform pricing — surfaces non-obvious price-structure choices.

WHEN IT IS USED

in regulatory analysis — competition authorities now treat two-sided pricing as standard.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Platform Economics.

REAL-WORLD EXAMPLES

● Google Search: free for users; advertisers pay per click.

● Adobe Acrobat: Reader free; full Pro paid.

● Credit cards: cardholders often paid (rewards); merchants pay interchange.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Subsidised Side Pricing contributes a clearer, more transferable framing of the dynamic it names — Charge one side · subsidise the other · maximise platform-level surplus. Its analytical contribution is the explicit recognition that side that contributes more to the OTHER side's value — subsidise.

RELATED FRAMEWORKS IN THIS COURSE

Chicken-and-Egg Problem · 3.3.1

Two-Sided Markets — Market Structure View · 3.1.2

Freemium Model (Unit 2.3.2)

SOURCES (HARVARD REFERENCING STYLE)

Armstrong, M. (2006) 'Competition in two-sided markets', RAND Journal of Economics, 37(3), pp. 668–691.

Evans, D.S. and Schmalensee, R. (2016) Matchmakers: The New Economics of Multisided Platforms. Boston: HBR Press.

Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', Journal of the European Economic Association, 1(4), pp. 990–1029.

Platform Governance

SLIBRARY 11 · SHEET
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The platform is also the rule-maker — admission, conduct, dispute, exit.

PROFESSIONAL DEFINITION

The set of rules — formal and algorithmic — by which a platform regulates the conduct of its users, complementors, and partners. Encompasses access criteria, content standards, ranking algorithms, dispute resolution, and exit rules. Increasingly the subject of regulation (DSA, P2B Regulation).

WHO DEVELOPED IT

Boudreau, K.J. and Hagiu, A. (2009) 'Platform Rules: Multi-Sided Platforms as Regulators', in Gawer (ed.) 'Platforms, Markets and Innovation'.

WHY IT WAS DEVELOPED

Klonick, K. (2018) 'The New Governors', Harvard Law Review, applied this lens to content-moderation governance.

FIGURE 1 · PLATFORM GOVERNANCE



Figure 1. Platform Governance: The platform is also the rule-maker — admission, conduct, dispute, exit.

HOW IT IS USED

when designing a new platform — governance must be designed before launch, not retrofitted.

WHEN IT IS USED

in legal and reputational risk assessment.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Platform Economics.

REAL-WORLD EXAMPLES

- App Store rules: enforced unilaterally; subject of DMA enforcement.
- YouTube Community Guidelines: algorithmic + human-moderation hybrid.
- Twitter/X policy reversals (2022–): governance volatility eroded user trust.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Platform Governance contributes a clearer, more transferable framing of the dynamic it names — The platform is also the rule-maker — admission, conduct, dispute, exit. Its analytical contribution is the explicit recognition that four governance dimensions: (i) Access — who can join; (ii) Admission of content / activity — what is allowed; (iii) Engagement — how the platform shapes interactions through ranking and recommendation; (iv) Exit — who is removed and how.

RELATED FRAMEWORKS IN THIS COURSE

DSA · 3.4.2

Multi-Sided Platforms · 3.1.3

Algorithmic Bias (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Boudreau, K.J. and Hagiu, A. (2009) 'Platform rules: multi-sided platforms as regulators', in Gawer, A. (ed.) Platforms, Markets and Innovation. Cheltenham: Edward Elgar, pp. 163–191.

Klonick, K. (2018) 'The new governors: the people, rules, and processes governing online speech', Harvard Law Review, 131(6), pp. 1598–1670.

Gillespie, T. (2018) Custodians of the Internet. New Haven: Yale University Press.

Switching Costs in Platforms

SLIBRARY 11 · SHEET
4 / 5

The frictions that lock users in — and the regulatory tools to break them.

PROFESSIONAL DEFINITION

The technical, contractual, social, and psychological costs that users incur when migrating between platforms. From the platform's perspective, switching costs underpin pricing power and customer lifetime value; from the regulator's perspective, they are a primary target of data-portability and interoperability mandates.

WHO DEVELOPED IT

Klemperer, P. (1995) 'Competition When Consumers Have Switching Costs', Review of Economic Studies.

WHY IT WAS DEVELOPED

Refined for digital by Shapiro & Varian (1999) and again by EU Digital Markets Act (2022).

FIGURE 1 · SWITCHING COSTS IN PLATFORMS

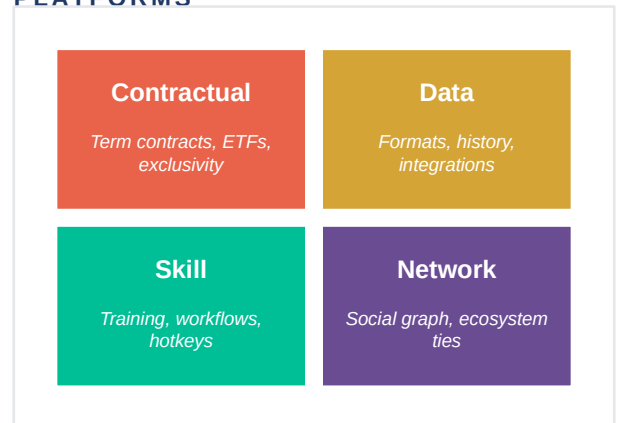


Figure 1. Switching Costs in Platforms: The frictions that lock users in — and the regulatory tools to break them.

HOW IT IS USED

when assessing the durability of a platform's competitive position.

WHEN IT IS USED

when designing data-portability and interoperability features required by the DMA.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Platform Economics.

REAL-WORLD EXAMPLES

- WhatsApp: social-graph lock-in — DMA forces interoperability with messaging from 2024.
- Apple iCloud: format and integration lock-in — partial data-portability under EU pressure.
- AWS: data-egress fees and bespoke services (Lambda, DynamoDB) — data + skill lock-in.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Switching Costs in Platforms contributes a clearer, more transferable framing of the dynamic it names — The frictions that lock users in — and the regulatory tools to break them. Its analytical contribution is the explicit recognition that four types: (i) Contractual — long-term contracts, exit fees, tied subscriptions; (ii) Data — formats, history, integrations that don't transfer; (iii) Skill — training, certifications, learned workflows; (iv) Network — social graph, group ties, reputation that lives on the platform.

RELATED FRAMEWORKS IN THIS COURSE

[Lock-In & Switching Costs \(Unit 2.4.5\)](#)
[Platform Governance · 3.3.3](#)
[DMA · 3.4.1](#)

SOURCES (HARVARD REFERENCING STYLE)

Klemperer, P. (1995) 'Competition when consumers have switching costs', Review of Economic Studies, 62(4), pp. 515–539.

Shapiro, C. and Varian, H.R. (1999) Information Rules: A Strategic Guide to the Network Economy. Boston: HBS Press.

European Union (2022) Regulation (EU) 2022/1925 — Digital Markets Act. Brussels: Official Journal of the EU.

Data Economies of Scale

SLIBRARY 11 · SHEET
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More users → more data → better product → more users — the data flywheel.

PROFESSIONAL DEFINITION

A feedback loop in which scale generates data, data improves the product, and the improved product attracts more users. Distinct from network effects: the gain comes from data volume, not from interactions among users. Particularly important in AI-driven products where model quality depends on training data.

WHO DEVELOPED IT

Term in industry use since ~2015.

WHY IT WAS DEVELOPED

Academic foundation in Hagiu, A. and Wright, J. (2020) 'When Data Creates Competitive Advantage', Harvard Business Review. UK CMA (2020) 'Online Platforms and Digital Advertising' provides empirical case.

FIGURE 1 · DATA ECONOMIES OF SCALE

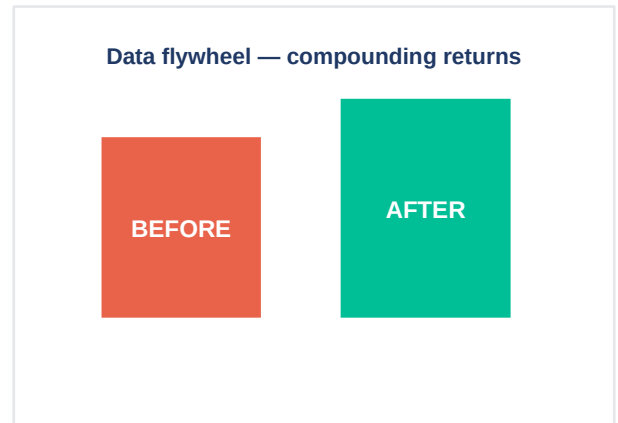


Figure 1. Data Economies of Scale: More users → more data → better product → more users — the data flywheel.

HOW IT IS USED

when assessing competitive moats in AI-intensive products.

WHEN IT IS USED

in M&A diligence — data assets often dominate post-acquisition synergy thesis.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Platform Economics.

REAL-WORLD EXAMPLES

● Google Search: query data → better ranking → more usefulness → retention.

● Tesla Autopilot: 5bn+ miles of driving data; OTA updates redistribute gains.

● OpenAI ChatGPT: RLHF on user feedback continuously refines model quality.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Data Economies of Scale contributes a clearer, more transferable framing of the dynamic it names — More users → more data → better product → more users — the data flywheel. Its analytical contribution is the explicit recognition that four conditions for a data economy of scale: (i) more usage → more data; (ii) more data → better product (non-trivially); (iii) better product → more users; (iv) data must be defensible (not commoditised).

RELATED FRAMEWORKS IN THIS COURSE

Data as Strategic Asset (Unit 2.3.4)

AI Spectrum (Unit 4)

DMA · 3.4.1

SOURCES (HARVARD REFERENCING STYLE)

Hagiu, A. and Wright, J. (2020) 'When data creates competitive advantage', Harvard Business Review, 98(1), pp. 94–101.

UK Competition and Markets Authority (2020) Online Platforms and Digital Advertising — Final Report. London: CMA.

Bajari, P., Chernozhukov, V., Hortaçsu, A. and Suzuki, J. (2019) 'The impact of big data on firm performance', NBER Working Paper 24334.

DMA — Digital Markets Act

EU Regulation 2022/1925 — ex-ante rules for designated 'gatekeeper' platforms.

PROFESSIONAL DEFINITION

EU Regulation 2022/1925 establishes obligations on designated 'gatekeeper' platforms — large core-platform-services providers that act as important gateways between business users and end users. The DMA imposes ex-ante rules (banned conduct, mandatory access, interoperability) without requiring case-by-case enforcement.

WHO DEVELOPED IT

Drafted by DG-COMP and DG-CNECT 2020–2022; Regulation (EU) 2022/1925 entered into force 2 May 2023; main obligations applied from 7 March 2024.

WHY IT WAS DEVELOPED

Intellectual roots in EU competition law (Article 102 TFEU) and the EC's Vestager-era Google cases.

FIGURE 1 · DMA — DIGITAL MARKETS ACT

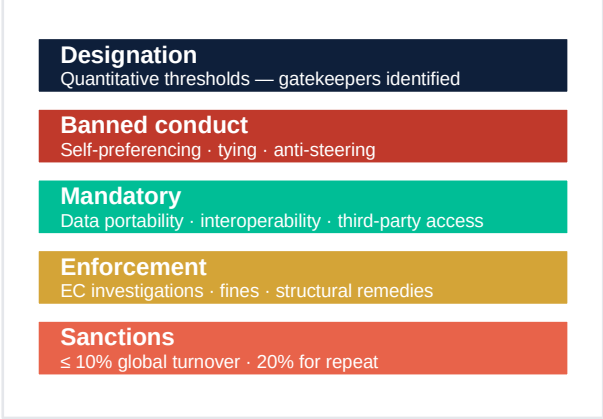


Figure 1. DMA — Digital Markets Act: EU Regulation 2022/1925 — ex-ante rules for designated 'gatekeeper' platforms.

HOW IT IS USED

when assessing compliance obligations for any large platform with EU presence.

WHEN IT IS USED

in competitive analysis — DMA reshapes the market structure of digital platforms.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Failures & Regulation.

REAL-WORLD EXAMPLES

- Designated gatekeepers (2023–24): Alphabet, Amazon, Apple, ByteDance, Meta, Microsoft.
- First Apple App Store fine: €1.84bn for anti-steering (March 2024).
- Google Search self-preferencing investigations ongoing 2024–2025.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, DMA — Digital Markets Act contributes a clearer, more transferable framing of the dynamic it names — EU Regulation 2022/1925 — ex-ante rules for designated 'gatekeeper' platforms. Its analytical contribution is the explicit recognition that designation: firms above thresholds (€7.5bn EU turnover or €75bn market cap, plus 45m EU monthly users on a core platform service) are designated gatekeepers.

RELATED FRAMEWORKS IN THIS COURSE

- DSA · 3.4.2
- Antitrust in Digital Markets · 3.4.3
- GAFAM Concentration · 3.1.5

SOURCES (HARVARD REFERENCING STYLE)

European Union (2022) Regulation (EU) 2022/1925 — Digital Markets Act. Brussels: Official Journal of the EU.

Cabral, L., Haucap, J., Parker, G., Petropoulos, G., Valletti, T. and Van Alstyne, M. (2021) The EU Digital Markets Act: A Report from a Panel of Economic Experts. Luxembourg: Publications Office of the EU.

European Commission (2024) Designating Gatekeepers under the Digital Markets Act. Brussels: European Commission.

DSA — Digital Services Act

EU Regulation 2022/2065 — tiered obligations on online intermediaries.

PROFESSIONAL DEFINITION

EU Regulation 2022/2065 establishes harmonised obligations on online intermediaries operating in the EU. Obligations scale with size: minimal for all hosting providers, more for platforms, most for Very Large Online Platforms (VLOPs) and Very Large Online Search Engines (VLOSEs) reaching 45m+ monthly EU users.

WHO DEVELOPED IT

Replaced parts of the e-Commerce Directive 2000/31/EC.

WHY IT WAS DEVELOPED

Drafted alongside the DMA 2020–2022; entered into force 16 November 2022; full applicability for VLOPs/VLOSEs from August 2023, all others from February 2024. Intellectual roots: e-Commerce Directive's safe-harbour regime; Klonick (2018) 'New Governors'.

FIGURE 1 · DSA — DIGITAL SERVICES ACT

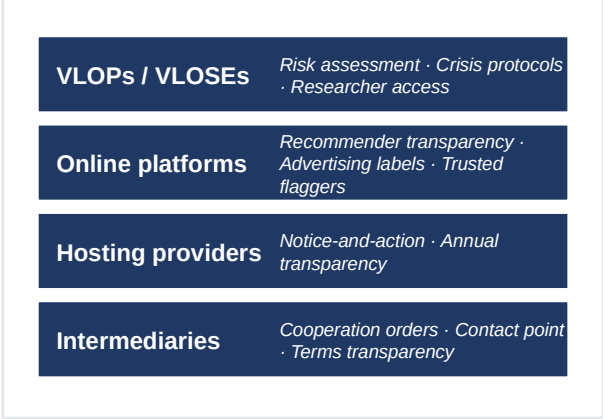


Figure 1. DSA — Digital Services Act: EU Regulation 2022/2065 — tiered obligations on online intermediaries.

HOW IT IS USED

when designing trust-and-safety operations on any EU-facing online service.

WHEN IT IS USED

in regulatory and reputational risk assessment.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Failures & Regulation.

REAL-WORLD EXAMPLES

- First 19 VLOPs/VLOSEs designated April 2023: Amazon, Apple, Meta, Google, X, TikTok, others.
- First DSA penalty proceedings: X/Twitter (2024) — illegal content and dark patterns.
- TikTok DSA proceedings (2024): minor protection, addictive design.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, DSA — Digital Services Act contributes a clearer, more transferable framing of the dynamic it names — EU Regulation 2022/2065 — tiered obligations on online intermediaries. Its analytical contribution is the explicit recognition that four tiers of obligation, scaling with size and risk: Intermediaries → Hosting → Platforms → VLOPs / VLOSEs.

RELATED FRAMEWORKS IN THIS COURSE

- DMA · 3.4.1
- Platform Governance · 3.3.3
- GDPR Principles (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

- European Union (2022) Regulation (EU) 2022/2065 — Digital Services Act. Brussels: Official Journal of the EU.
- Husovec, M. (2024) Principles of the Digital Services Act. Oxford: Oxford University Press.
- European Commission (2023) DSA: Designation of VLOPs and VLOSEs. Brussels: European Commission.

Antitrust in Digital Markets

SLIBRARY 12 · SHEET
3 / 5

Article 102 TFEU · Sherman Act · all GAFAM firms under active case as of 2025.

PROFESSIONAL DEFINITION

The application of competition law — Article 102 TFEU in the EU; Section 2 of the Sherman Act in the US — to the conduct and structure of digital firms. As of 2025, every GAFAM firm is the subject of at least one major competition case in the US, EU, UK, or other jurisdictions.

WHO DEVELOPED IT

Modern digital antitrust era opens with EC Microsoft case (2004).

WHY IT WAS DEVELOPED

Lina Khan's 'Amazon's Antitrust Paradox' (2017, Yale Law Journal) reframed academic and policy debate. EC Vestager-era Google cases (2017–2019) established multi-billion-euro penalties.

FIGURE 1 · ANTITRUST IN DIGITAL MARKETS



Figure 1. Antitrust in Digital Markets: Article 102 TFEU · Sherman Act · all GAFAM firms under active case as of 2025.

HOW IT IS USED

when designing platform business strategy with regulatory tailwinds in mind.

WHEN IT IS USED

in competitive due diligence — antitrust outcomes reshape competitive landscape.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Market Failures & Regulation.

REAL-WORLD EXAMPLES

- US v. Google Search (2024 ruling, remedies 2025): adjudicated illegal monopoly.
- US v. Apple App Store (2024, ongoing): bundling and revenue share.
- FTC v. Meta (2020+, ongoing): Instagram and WhatsApp acquisitions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Antitrust in Digital Markets contributes a clearer, more transferable framing of the dynamic it names — Article 102 TFEU · Sherman Act · all GAFAM firms under active case as of 2025. Its analytical contribution is the explicit recognition that four enforcement modes: (i) Conduct cases — abuse of dominance (self-preferencing, tying); (ii) Merger review — blocking acquisitions of nascent competitors; (iii) Structural cases — divestiture remedies; (iv) Market investigations — UK CMA market studies.

RELATED FRAMEWORKS IN THIS COURSE

DMA · 3.4.1

GAFAM Concentration · 3.1.5

Essential Facilities Doctrine · 3.4.5

SOURCES (HARVARD REFERENCING STYLE)

- Khan, L. (2017) 'Amazon's antitrust paradox', Yale Law Journal, 126(3), pp. 710–805.
- Crémer, J., de Montjoye, Y.-A. and Schweitzer, H. (2019) Competition Policy for the Digital Era. Brussels: European Commission.
- Furman, J. et al. (2019) Unlocking Digital Competition: Report of the Digital Competition Expert Panel. London: HM Government.

Algorithmic Collusion

SLIBRARY 12 · SHEET
4 / 5

Pricing algorithms learning to cooperate without communicating — a market failure for the AI era.

PROFESSIONAL DEFINITION

Algorithmic collusion occurs when independent pricing algorithms — typically reinforcement-learning agents — learn to cooperate on supra-competitive prices without explicit agreement among the firms that deploy them. The classic competition-law requirement of 'meeting of minds' fails to apply, creating an enforcement gap.

WHO DEVELOPED IT

Theoretical demonstration in Calvano, E., Calzolari, G., Denicolò, V. and Pastorello, S. (2020) 'Artificial intelligence, algorithmic pricing, and collusion', American Economic Review.

WHY IT WAS DEVELOPED

Earlier policy alarm: Ezrachi & Stucke (2016) 'Virtual Competition'.

FIGURE 1 · ALGORITHMIC COLLUSION

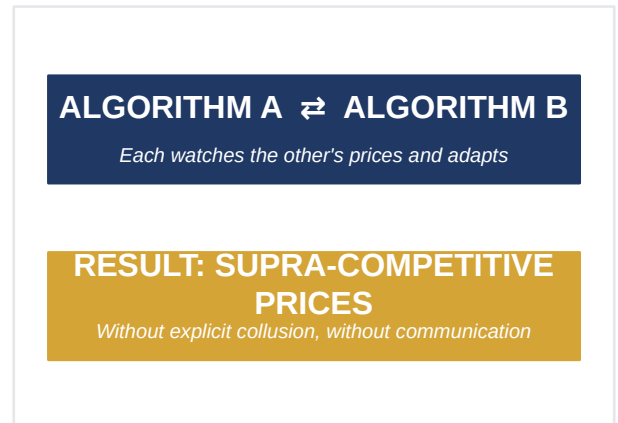


Figure 1. Algorithmic Collusion: Pricing algorithms learning to cooperate without communicating — a market failure for the AI era.

HOW IT IS USED

when assessing competition-law risk in algorithmic-pricing systems.

WHEN IT IS USED

in regulatory and antitrust analysis of AI-driven markets.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Market Failures & Regulation.

REAL-WORLD EXAMPLES

- RealPage rental-pricing case (US 2022+): DOJ and several states allege algorithmic price-fixing among landlords.
- Topkins (US, 2015): rare conviction for algorithmic price-fixing in poster-print sales.
- Empirical study in retail petrol (Germany): margin compression slowed after rollout of algorithmic pricing software.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Algorithmic Collusion contributes a clearer, more transferable framing of the dynamic it names — Pricing algorithms learning to cooperate without communicating — a market failure for the AI era. Its analytical contribution is the explicit recognition that pricing algorithms observe rival behaviour and adjust own prices.

RELATED FRAMEWORKS IN THIS COURSE

Antitrust in Digital Markets · 3.4.3

Algorithmic Bias (Unit 5).

Platform Governance · 3.3.3

SOURCES (HARVARD REFERENCING STYLE)

Calvano, E., Calzolari, G., Denicolò, V. and Pastorello, S. (2020) 'Artificial intelligence, algorithmic pricing, and collusion', American Economic Review, 110(10), pp. 3267–3297.

Ezrachi, A. and Stucke, M.E. (2016) Virtual Competition. Cambridge, MA: Harvard University Press.

OECD (2017) Algorithms and Collusion: Competition Policy in the Digital Age. Paris: OECD Publishing.

Essential Facilities Doctrine

SLIBRARY 12 · SHEET
5 / 5

When access to a bottleneck input is a precondition for competition.

PROFESSIONAL DEFINITION

The competition-law doctrine that a dominant firm controlling an 'essential facility' — an input that downstream competitors cannot reasonably duplicate — must grant access on fair, reasonable, and non-discriminatory (FRAND) terms. Originating in 19th-century US railway cases; revived for digital platforms in EU debate.

WHO DEVELOPED IT

US origin: Terminal Railroad Association case (US, 1912).

WHY IT WAS DEVELOPED

EU formalisation: Bronner case (Case C-7/97, 1998 ECJ).

Modern application to digital platforms: Magill (1995) and IMS Health (2004) for IP-as-essential-facility.

FIGURE 1 · ESSENTIAL FACILITIES DOCTRINE



Figure 1. Essential Facilities Doctrine: When access to a bottleneck input is a precondition for competition.

HOW IT IS USED

when assessing whether a digital input qualifies for mandatory-access remedies.

WHEN IT IS USED

in policy analysis of platform regulation.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Market Failures & Regulation.

REAL-WORLD EXAMPLES

● Microsoft (2004): EC ordered Microsoft to license server-interoperability info.

● Google Shopping (2017+): self-preferencing ordered to stop.

● Apple App Store: subject of DMA mandatory-access obligations.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Essential Facilities Doctrine contributes a clearer, more transferable framing of the dynamic it names — When access to a bottleneck input is a precondition for competition. Its analytical contribution is the explicit recognition that three conditions to qualify (Bronner test): (i) refusal of access likely to eliminate all competition in the downstream market; (ii) the facility cannot be reasonably duplicated; (iii) refusal is not objectively justified.

RELATED FRAMEWORKS IN THIS COURSE

[Antitrust in Digital Markets · 3.4.3](#)[DMA · 3.4.1](#)[Switching Costs in Platforms · 3.3.4](#)

SOURCES (HARVARD REFERENCING STYLE)

European Court of Justice (1998) Case C-7/97 Oscar Bronner GmbH v. Mediaprint. Luxembourg: ECJ.

Lipsky, A.B. and Sidak, J.G. (1999) 'Essential facilities', Stanford Law Review, 51(5), pp. 1187–1248.

Petropoulos, G. (2024) 'Refusal to deal in digital markets: from essential facilities to interoperability', Yale Journal on Regulation, 41, pp. 1–62.